REPORT

Cat and Dog Image Classification using MobileNet and Flask API

INTODUCTION

The objective of this mini project was to develop a web application capable of classifying uploaded images as either a cat or a dog using a deep learning model. To achieve this, I used the MobileNet pre-trained architecture with TensorFlow/Keras for image classification and integrated the trained model into a Flask web application. The project provided practical experience in deep learning, transfer learning, model deployment, and web application development.

METHODOLOGY

The Cat and Dog dataset was preprocessed by resizing all images to 160 × 160 pixels and preparing them for training. A MobileNet-based model was trained using transfer learning, allowing the model to learn image features efficiently while reducing training time. After training, the model was saved in .keras format and used for prediction. A Flask API was then developed to accept image uploads from users. The uploaded image is processed using the same preprocessing steps as the training data, passed to the trained model, and classified as either Cat or Dog. The web interface displays the uploaded image, predicted class, and confidence score in a clean and user-friendly layout.

RESULTS

The developed application successfully classifies uploaded images with good accuracy. The integration of the trained MobileNet model with Flask enabled real-time predictions through a modern web interface. The project demonstrates how deep learning models can be deployed as practical web applications.

CONCLUSION

This project helped me understand the complete workflow of an image classification system, including data preprocessing, transfer learning with MobileNet, model training, evaluation, and deployment using Flask. It also improved my knowledge of integrating machine learning models into web applications and provided hands-on experience in building an end-to-end AI solution.